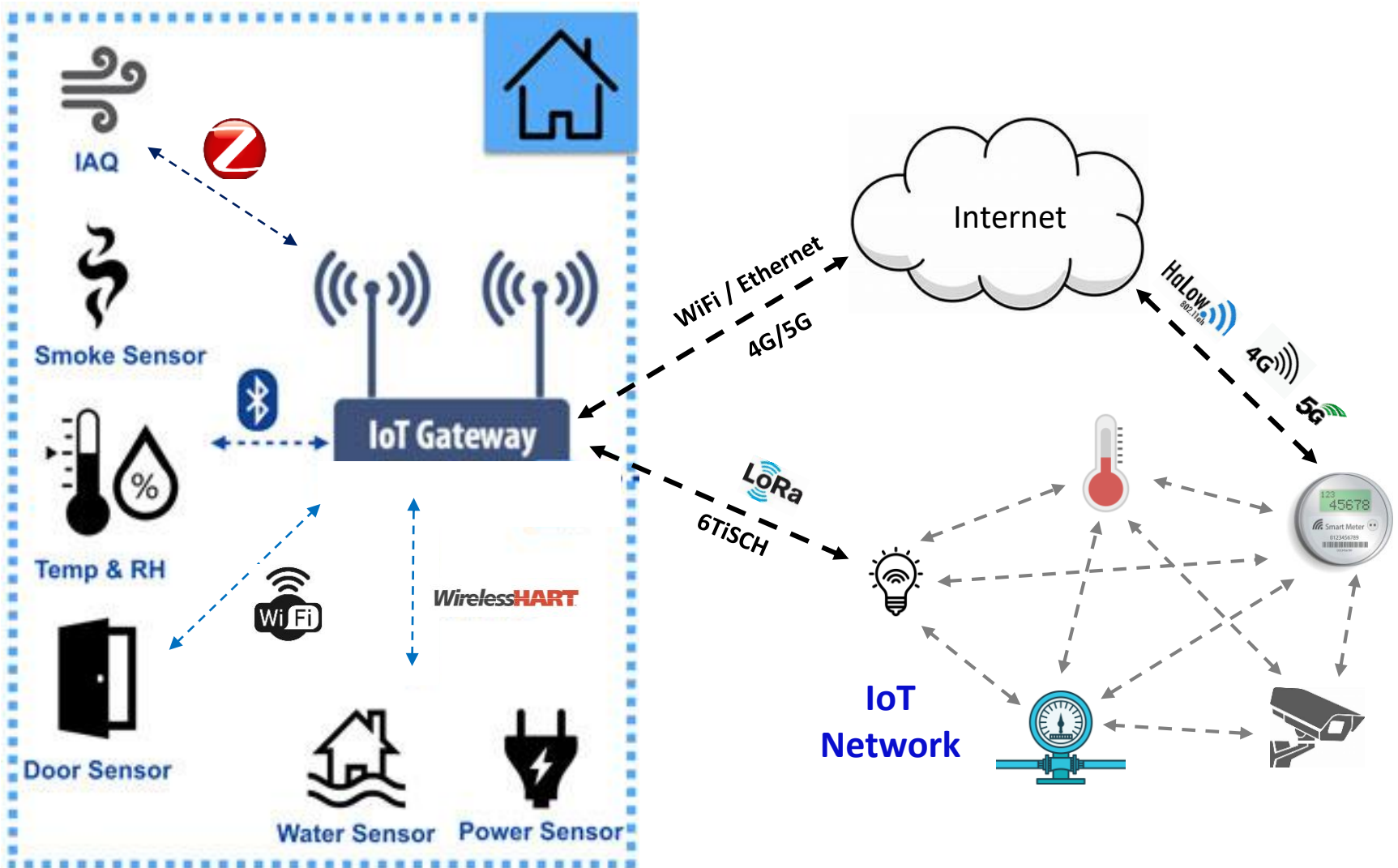


Internet of Things (IoT)



How IoT Works?

Smart Home IoT Application



Traditional Data Flow in IoT



Sensors collect real-world data.

Data is transmitted via local connectivity technologies.

Edge gateways aggregate and preprocess data.

Data is sent through the Internet.

Cloud performs storage and advanced processing.

Users access insights through applications.

Image Source: <http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf>

Traditional Data Flow in IoT

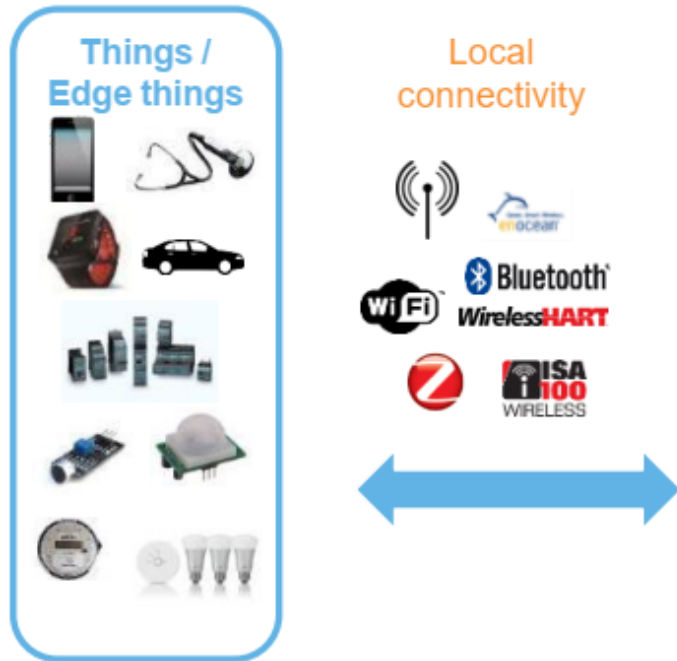


Image Source: <http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf>

Traditional Data Flow in IoT

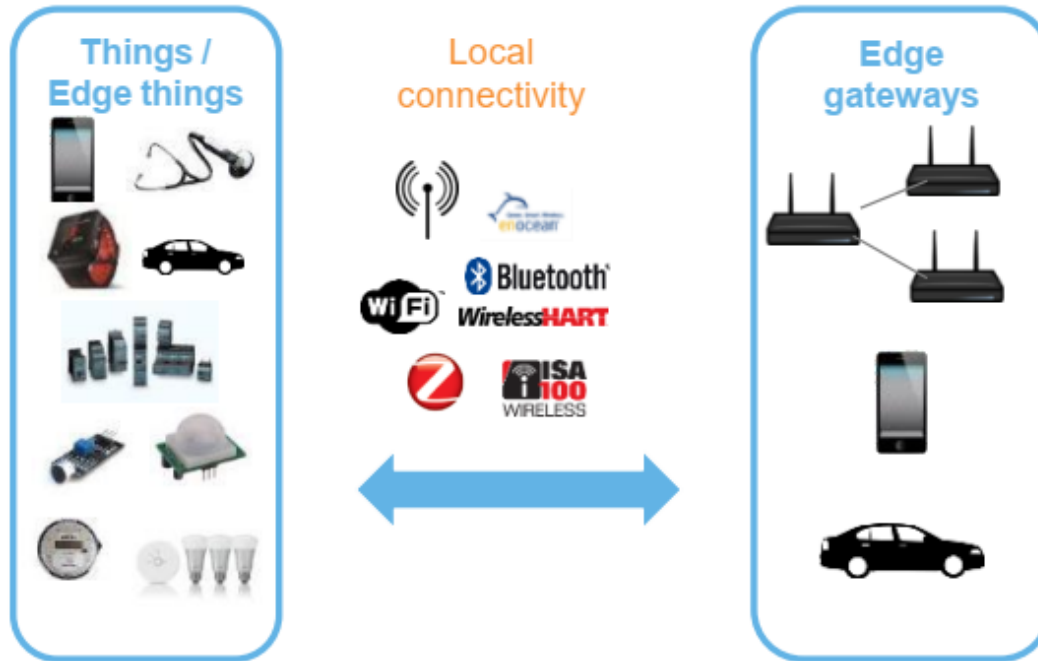


Image Source: <http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf>

Traditional Data Flow in IoT

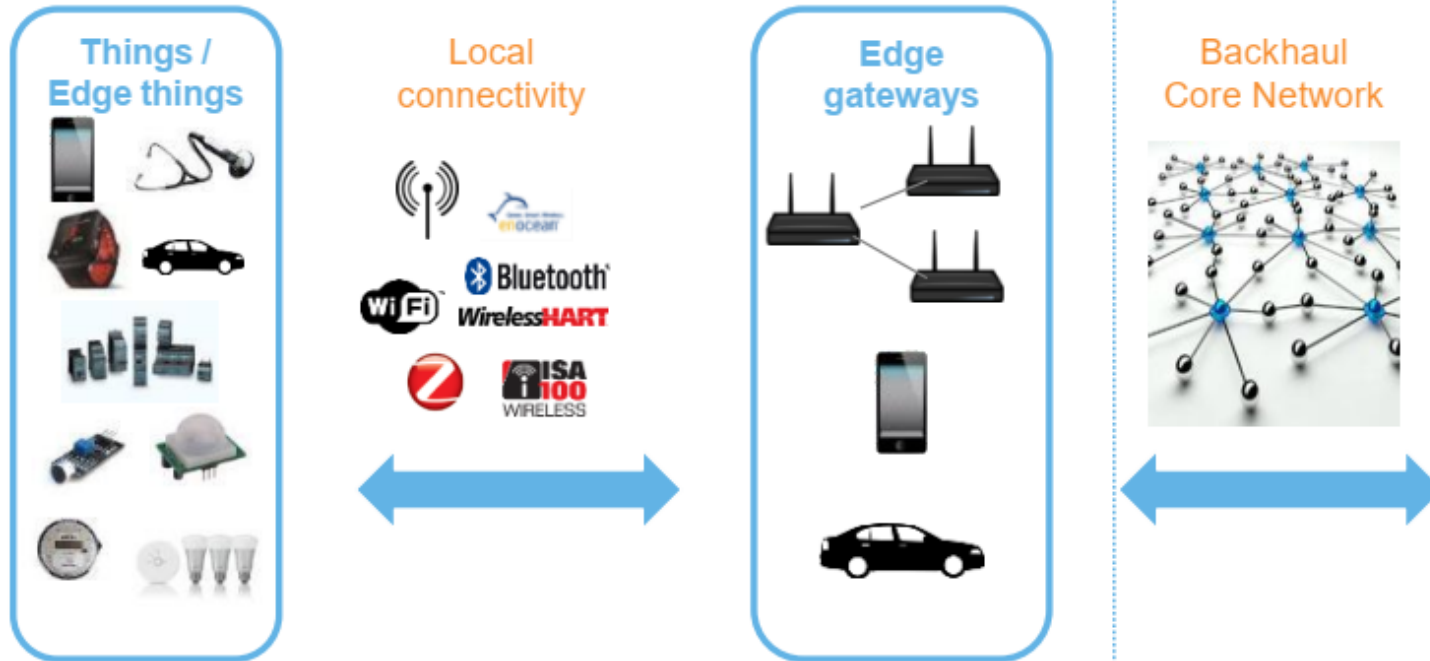


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Traditional Data Flow in IoT

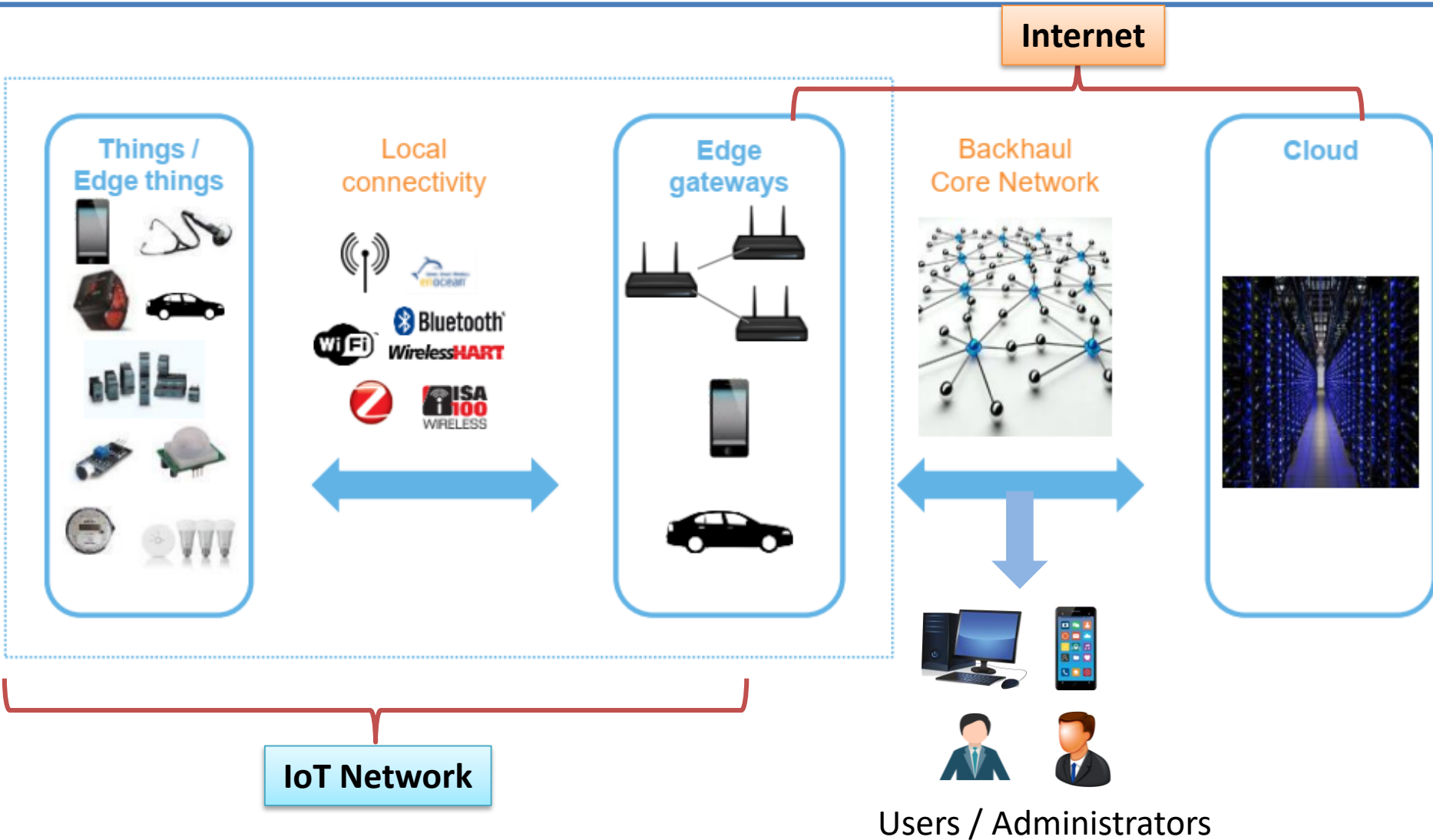
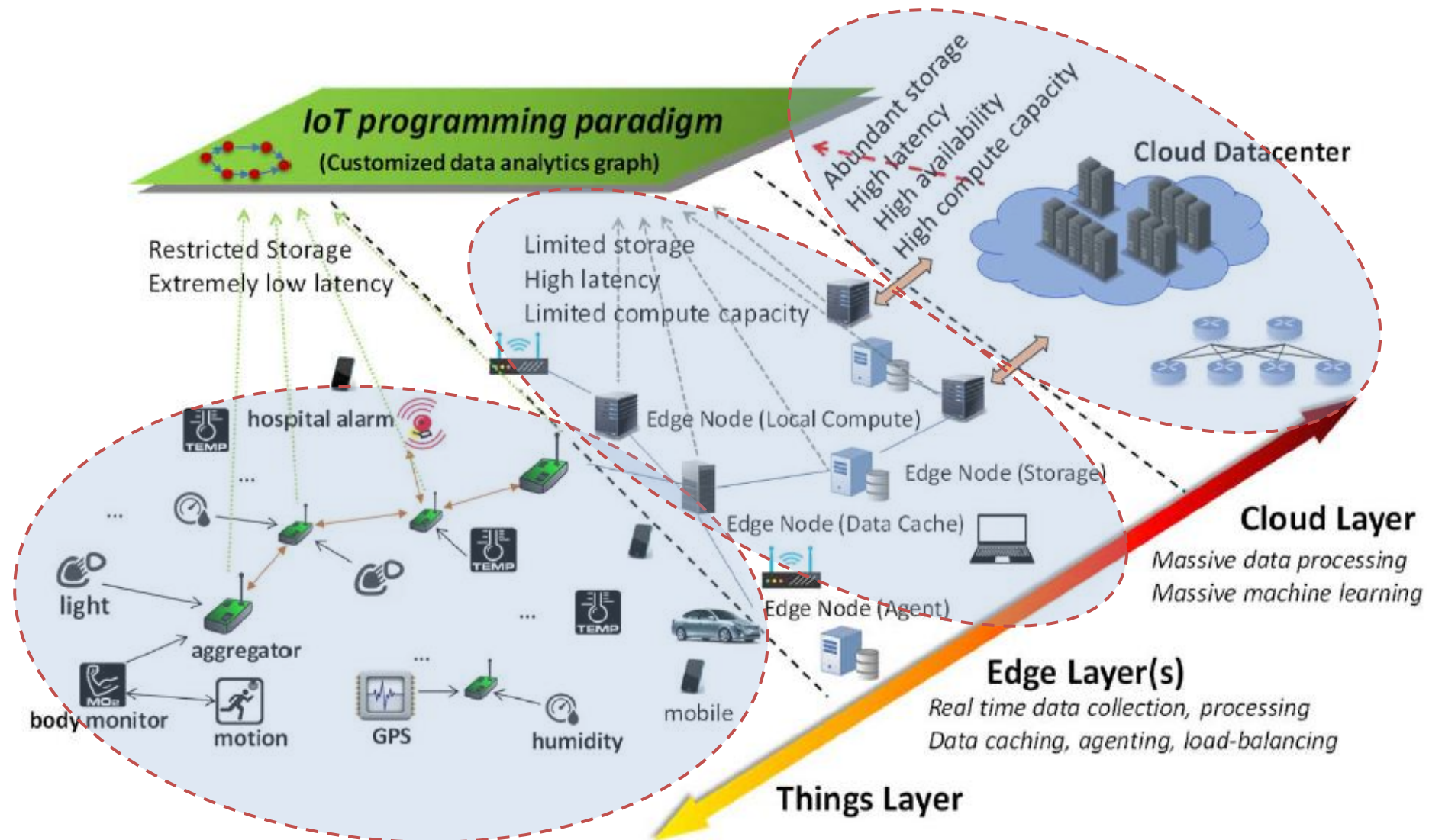


Image Source: <http://events17.linuxfoundation.org/sites/events/files/slides/Intelligence%20at%20the%20Edge.pdf>

Data Processing Locations

Immediate decisions → Things or Edge
Real-time analytics → Edge
Heavy AI training & big data → Cloud

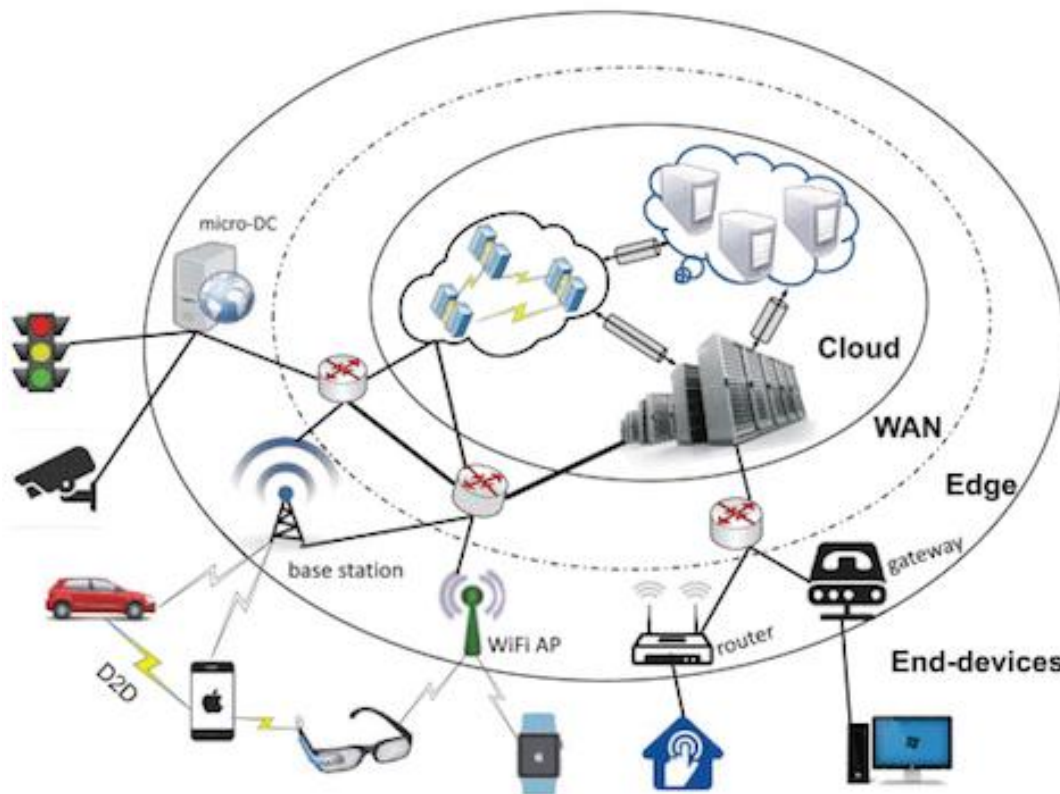


Source: Rajiv Ranjan et. al., "Integrating the IoT and Data Science" IEEE Cloud Computing, 2018

Data Processing at the Edge

Edge Computing

- Host computation tasks close to the data sources and end users.



AI:

- Extracts insights that leads to better decisions & strategic move

Edge Intelligence:

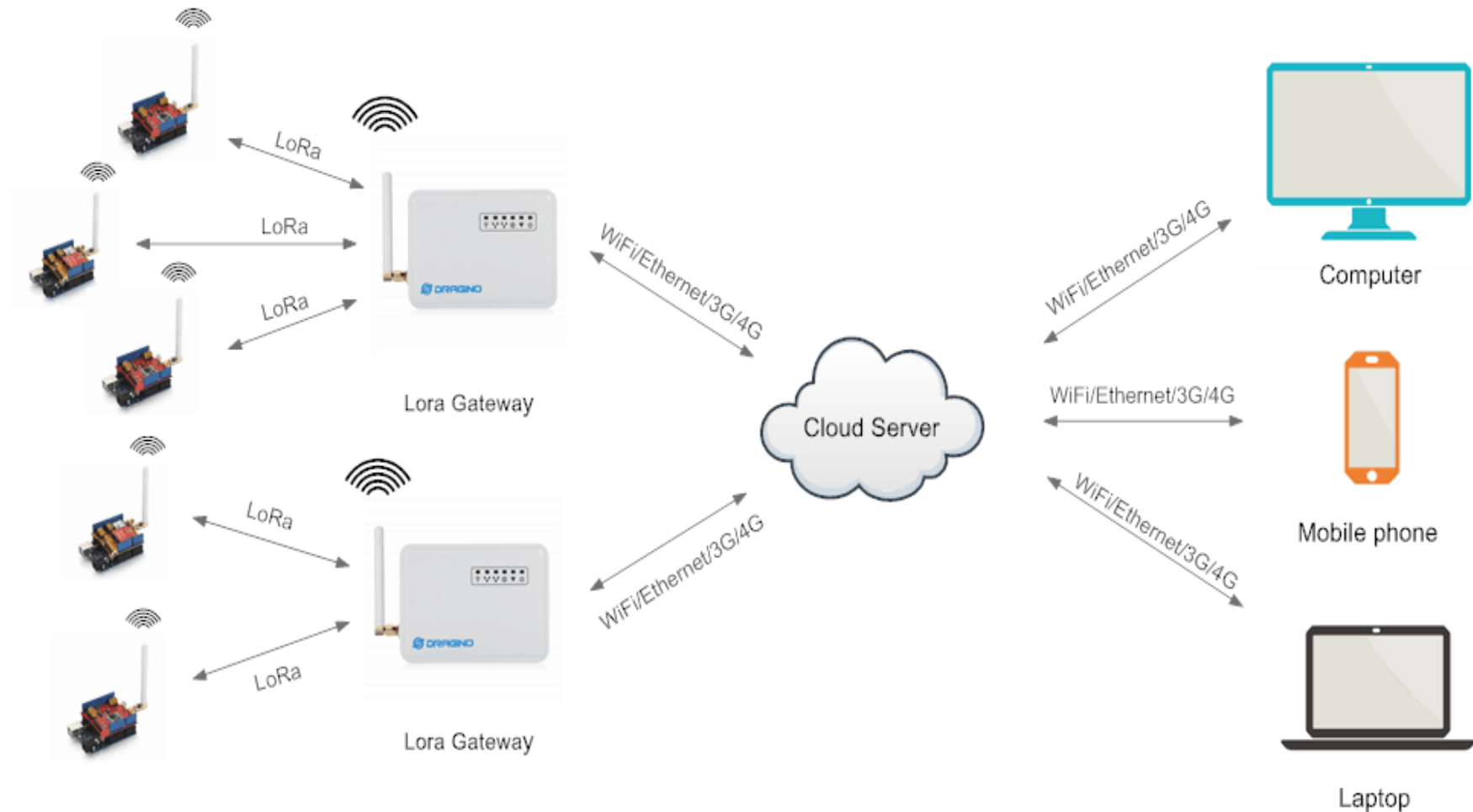
- Apply AI on data at network Edge

What is needed to know?

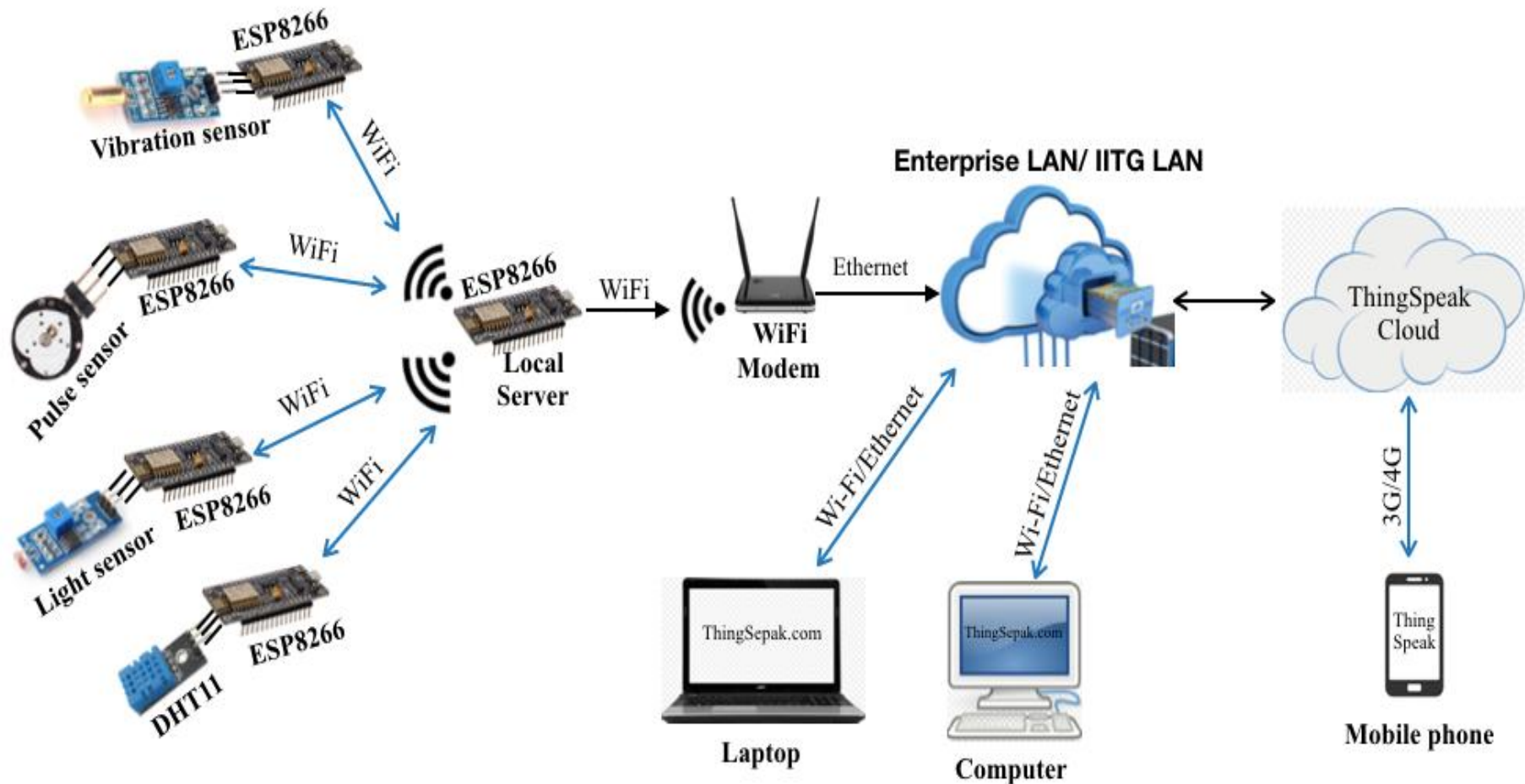


- **IoT Components**
 - Application developer needs to know what are the **required components** for designing an IoT application
- **IoT Architecture**
 - For any IoT application it is required to know the **physical arrangement** of IoT sensors, actuators, gateway, edge devices and Internet connectivity
- **IoT Network Access Technology**
 - Types of IoT Network Access Technology or **Connectivity Technology** used to connect the sensors/actuators with each other, and with the IoT gateway or Internet
- **IoT Data Processing / Analysis**
 - Sensed data needs to be **stored and analyzed** to take business decision or to get new business insight. Where it can be done and How?

DIY: Long-range Monitoring & Control



DIY: Smart Home Monitoring



Lessons Learned



- ✓ Layered View of an IoT Application
- ✓ Traditional Data Flow in IoT
- ✓ Data Processing Locations in IoT
- ✓ What is needed to know before developing IoT Applications

Thanks!

